

The background of the slide is a dense, repeating pattern of small, colorful fractal-like structures. These structures are primarily star-like or snowflake-like in shape, with some being circular and others more elongated. They are rendered in various colors including blue, green, orange, and purple, and are scattered across the entire slide area.

On the inverse volcano problem over non-prime finite fields

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December 11, 2025

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Charting the Landscape

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Elliptic curves and isogenies

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The *j -invariant* $j(E) \in \mathbb{F}_q$ uniquely characterises E up to isomorphism over $\overline{\mathbb{F}}_q$.

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An **isogeny** is a non-constant morphism $\varphi : E_1 \rightarrow E_2$. It is surjective with finite kernel.

- ▶ The **degree** of φ is the size of its kernel.¹
- ▶ Every isogeny $\varphi : E_1 \rightarrow E_2$ has a **dual isogeny** $\hat{\varphi} : E_2 \rightarrow E_1$ of the same degree.

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The **endomorphism ring** $\text{End}(E)$ consists of all isogenies $E \rightarrow E$ together with the zero constant map. Two cases:

- Ordinary**: $\text{End}(E)$ is an **order** $\mathcal{O}$ in an imaginary quadratic field K .
 - ▶ We say E has **complex multiplication** (CM) by $\mathcal{O}$.
- Supersingular**: $\text{End}(E)$ is a maximal order in a quaternion algebra over $\mathbb{Q}$.

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Isogeny graphs

Fix a prime ℓ coprime to q .

The ℓ -isogeny graph over $\mathbb{F}_q$ is the graph with:

- ▶ A vertex for each j -invariant in $\mathbb{F}_q$.
- ▶ A directed edge from $j(E_1)$ to $j(E_2)$ if there is an isogeny $\varphi : E_1 \rightarrow E_2$ of degree ℓ .¹

¹We identify two isogenies that are related by isomorphisms of the underlying elliptic curves.

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Two things of note:

- ▶ We identify an isogeny with its dual to make the graph essentially undirected (ignoring the two exceptional j -invariants 0 and 1728).
- ▶ Isogenies preserve the ordinary/supersingular distinction, so the graph can be partitioned into two parts.

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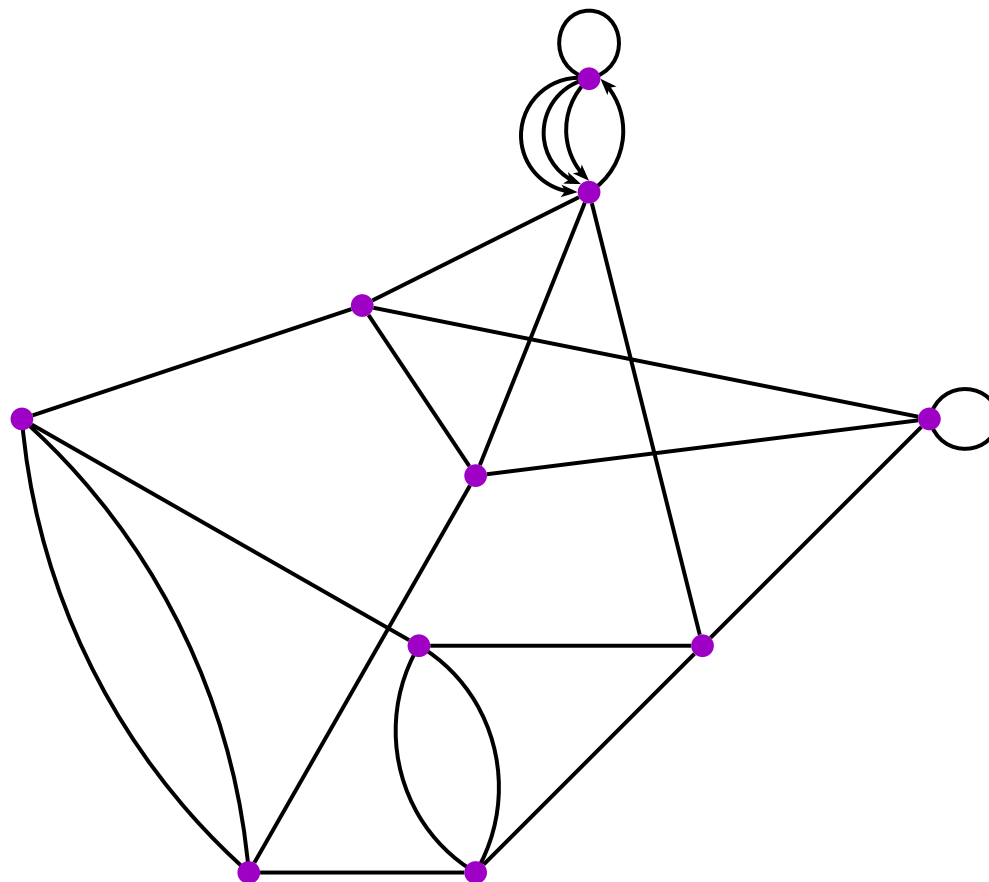
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- ▶ Isogenies preserve the ordinary/supersingular distinction, so the graph can be partitioned into two parts.

We are interested in the ordinary part, which we denote $\mathcal{G}_\ell(\mathbb{F}_q)$.

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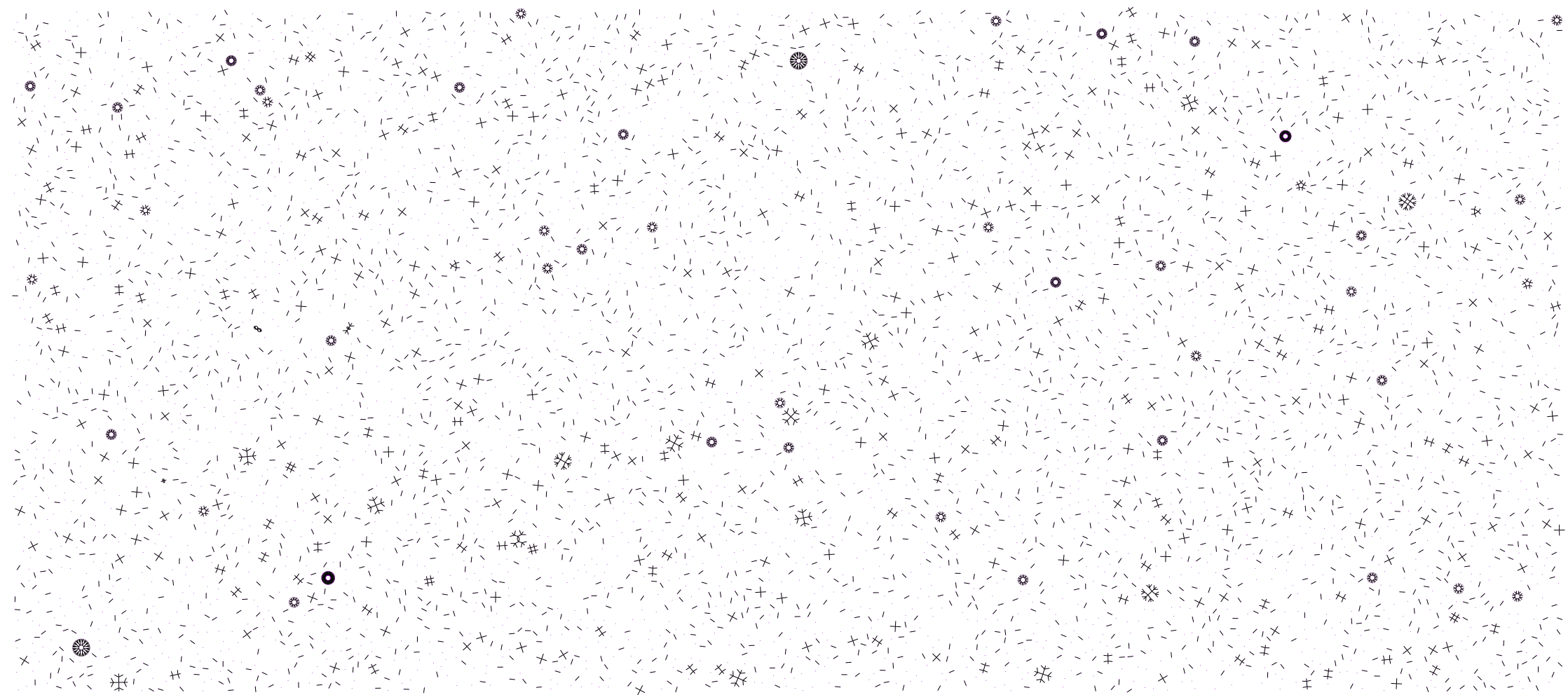
3-isogeny graph over $\mathbb{F}_{113^2}$: supersingular part

Connected and regular, but otherwise random-looking.



3-isogeny graph over $\mathbb{F}_{113^2}$: ordinary part

Disconnected, but components look (somewhat) structured?



Applications of isogeny graphs

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Cryptography (mainly supersingular):

- ▶ strong contenders for post-quantum cryptography protocols
- ▶ CGL hash function, SIKE key-exchange protocol, SQIsign digital signature protocol

See Goren and Love's survey paper [1] for an introduction to these things.

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Computation (mainly ordinary):

- ▶ determining whether a curve is supersingular
- ▶ computing the endomorphism ring of an ordinary curve
- ▶ computing Hilbert class polynomials
- ▶ computing modular polynomials

See Sutherland's expository paper [2] for examples of these algorithms.

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Ordinary isogeny graph's structure

Theorem (Kohel [3])

Let V be a component of the ordinary isogeny graph $\mathcal{G}_\ell(\mathbb{F}_q)$, and suppose V does not contain 0 or 1728. Then V is an ℓ -volcano¹.

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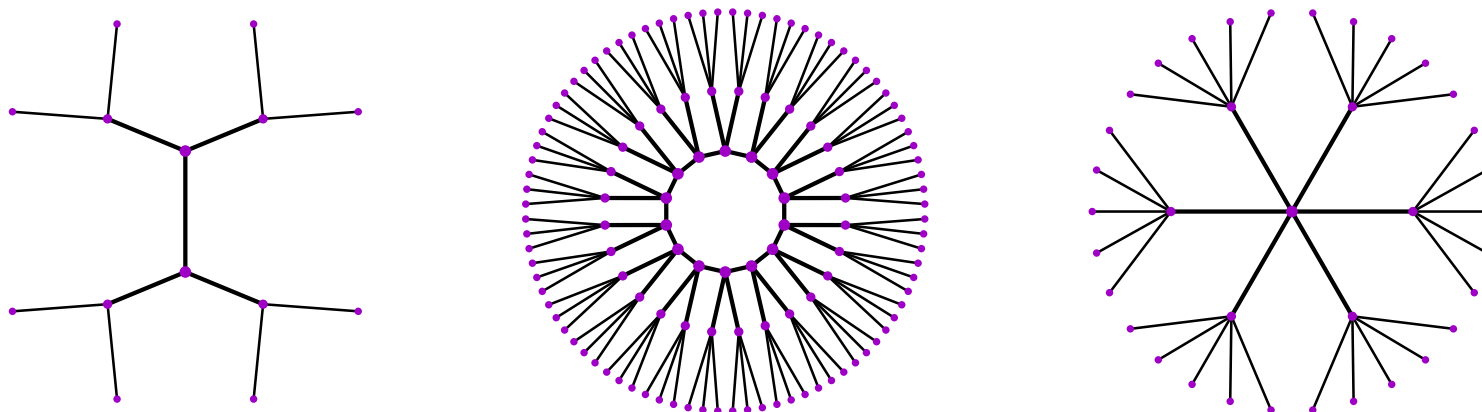


Figure: A 2-volcano, a 3-volcano, and a 5-volcano, respectively.

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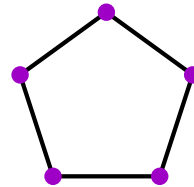
Anatomy of a volcano

Two parts:

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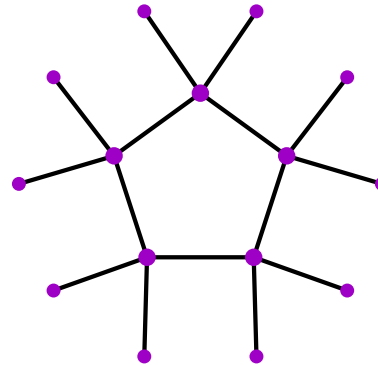
- ▶ the crater



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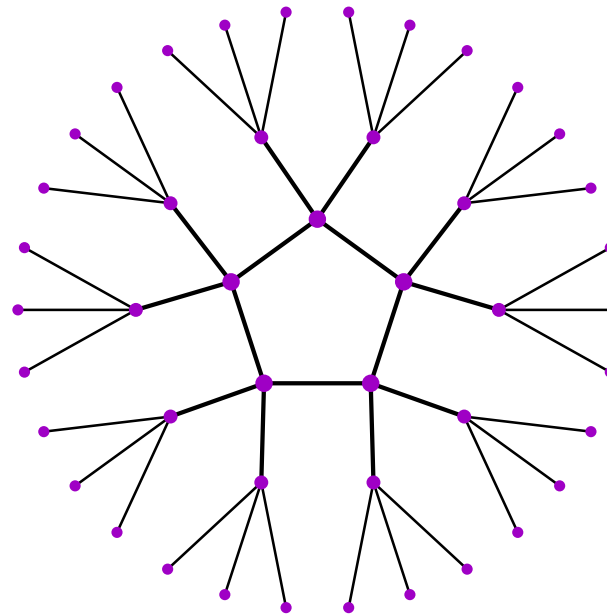
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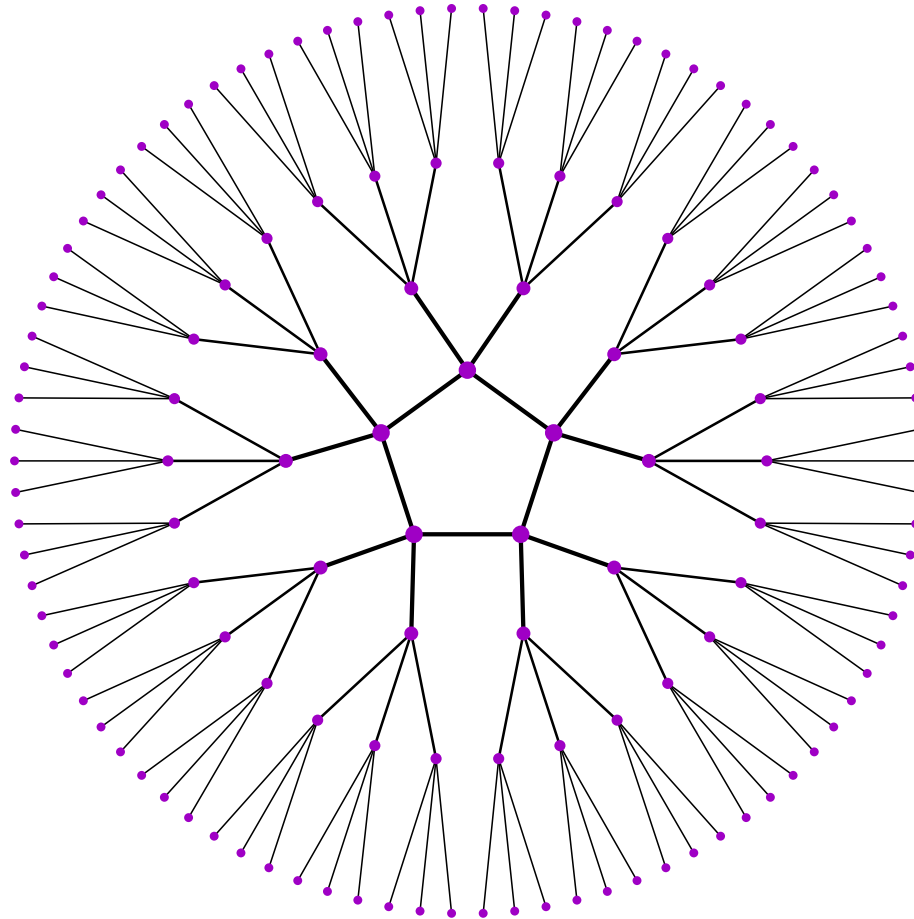
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Anatomy of a volcano

Two parts:

- ▶ the crater
- ▶ the lava flows



Anatomy of a volcano

More precisely, an ℓ -volcano of depth $d \geq 0$ is a connected undirected graph V whose vertices admit a partition into one or more levels $V_0, \dots, V_d$ such that

- ▶ the subgraph on V_0 (the crater) is a regular graph of degree at most¹ 2;
- ▶ for $0 \leq i < d$, each vertex in V_i has degree $\ell + 1$;
- ▶ for $0 < i \leq d$, each vertex in V_i has exactly one neighbour in level V_{i-1} .

¹Actually, this isn't entirely accurate...

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A volcano is completely determined by the 3-tuple (V_0, ℓ, d) .

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Anatomy of a volcano: the crater

A **crater** is a connected undirected graph V_0 that is either

- (a) a single vertex with zero, one, or two self-loops;
- (b) two vertices joined by one or two edges;
- (c) a cycle with ≥ 3 vertices.

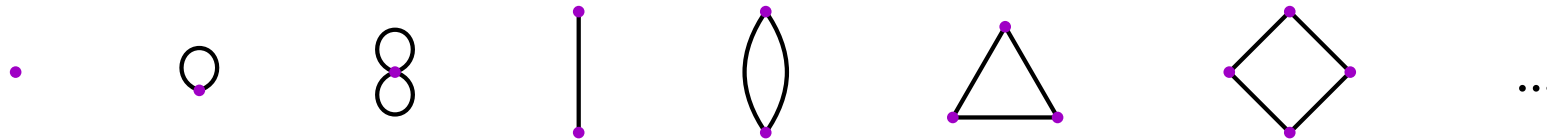


Figure: All possible craters.

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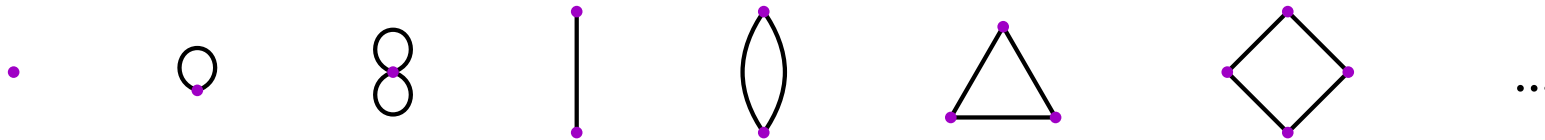


Figure: All possible craters.

Note:

- ▶ A volcano of depth 0 is just a crater.
- ▶ The crater is independent of ℓ .

Orders interlude

Let K be an imaginary quadratic field with discriminant D_K . An **order** is a subring $\mathcal{O}$ of finite index in the ring of integers $\mathcal{O}_K$.

- ▶ Has form $\mathcal{O} = \mathbb{Z} + c\mathcal{O}_K$ where $c = [\mathcal{O}_K : \mathcal{O}] \in \mathbb{Z}^+$ is called the **conductor** of $\mathcal{O}$.
- ▶ The discriminant of $\mathcal{O}$ is $D(\mathcal{O}) = c^2 D_K$.
- ▶ The **ideal class group** $\text{Cl}(\mathcal{O})$ is the group of invertible fractional $\mathcal{O}$ -ideals modulo the principal $\mathcal{O}$ -ideals.
- ▶ Class number $h(\mathcal{O}) = \# \text{Cl}(\mathcal{O})$.

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Given a prime ℓ ,¹ the ideal $\ell\mathcal{O}$ factors into prime $\mathcal{O}$ -ideals in one of three ways

- (a) **Inert**: $\ell\mathcal{O}$
- (b) **Ramified**: $\mathfrak{a}^2$
- (c) **Split**: $\mathfrak{a}\bar{\mathfrak{a}}$ with $\mathfrak{a} \neq \bar{\mathfrak{a}}$

¹We assume ℓ is coprime to the conductor c .

Orders interlude

The **Hilbert class field** H of K is the maximal unramified abelian extension.

- ▶ Key property: $\text{Gal}(H/K) \cong \text{Cl}(\mathcal{O}_K)$

The **ring class field** L of $\mathcal{O}$ generalises this to arbitrary orders.

- ▶ The maximal abelian extension unramified outside c such that a prime $\ell \nmid c$ splits completely iff ℓ is the norm of some $\alpha \in \mathcal{O}$.
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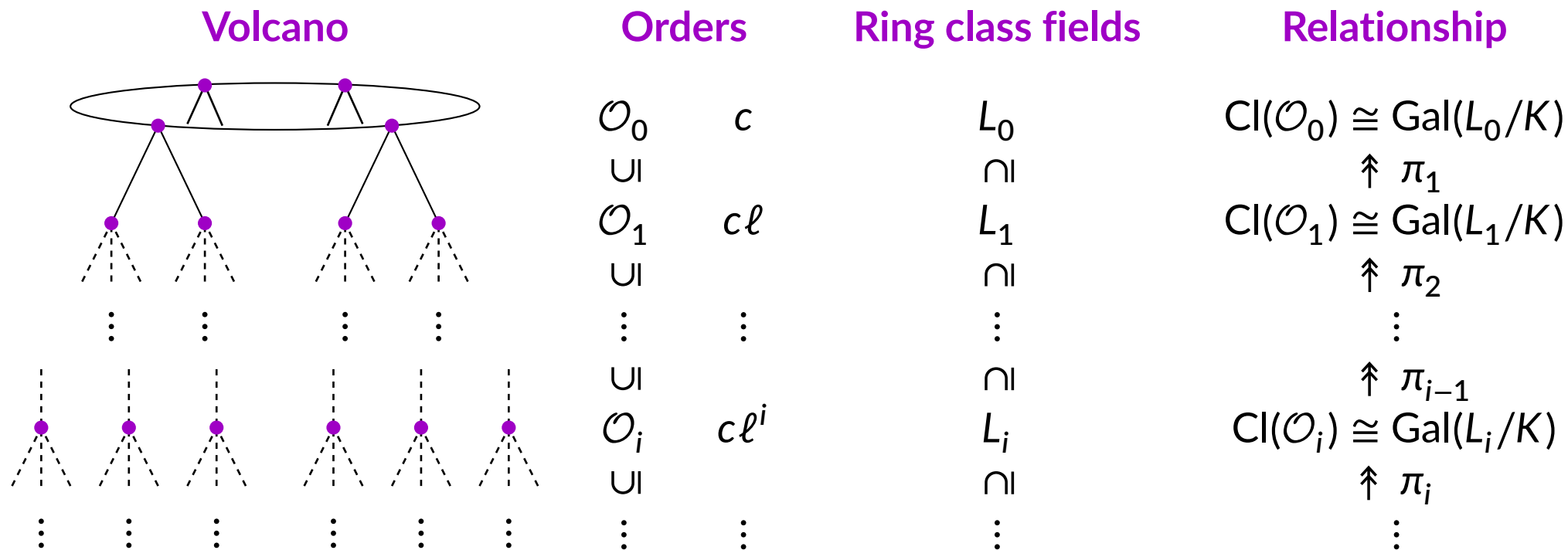
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We can relate $\mathcal{O}$ to $\mathcal{O}_K$ with the short exact sequence

$$\begin{array}{ccccccc}
 1 & \longrightarrow & \text{Gal}(L/H) & \longrightarrow & \text{Gal}(L/K) & \longrightarrow & \text{Gal}(H/K) \longrightarrow 1 \\
 & & & & \cong & & \cong \\
 & & & & \text{Cl}(\mathcal{O}) & & \text{Cl}(\mathcal{O}_K)
 \end{array}$$

The lava flows

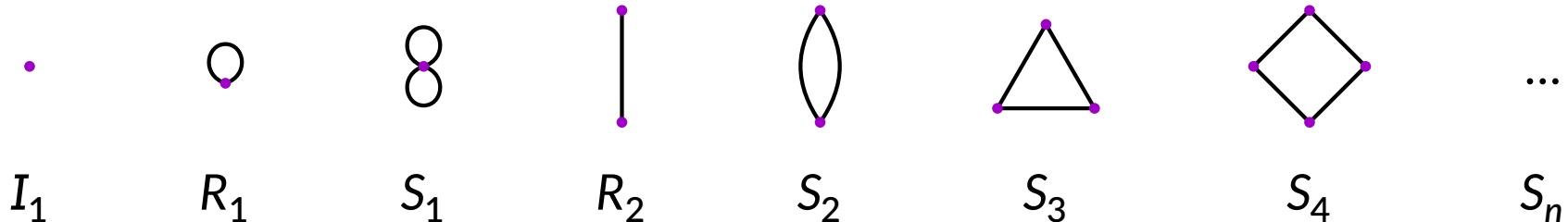


- ▶ All elliptic curves on the same level of a volcano have CM by the same order.
- ▶ The order at the crater has conductor coprime to ℓ .

The crater

Let V_0 be the crater of a volcano in $G_\ell(\mathbb{F}_q)$, let $\mathcal{O}$ be the endomorphism ring of any vertex in V_0 and $\mathfrak{a} \subseteq \mathcal{O}$ a prime above ℓ . Then V_0 is

- (I_1) a single vertex if ℓ is inert in $\mathcal{O}$
- (R_1) a single vertex with one loop if ℓ is ramified in $\mathcal{O}$ and $\mathfrak{a}$ is principal
- (S_1) a single vertex with two loops if ℓ is split in $\mathcal{O}$ and $\mathfrak{a}$ is principal
- (R_2) two vertices with a single edge if ℓ is ramified in $\mathcal{O}$ and $\mathfrak{a}$ is not principal
- (S_2) two vertices with two edges if ℓ is split in $\mathcal{O}$ and $[\mathfrak{a}]$ has order 2
- (S_n) an n -cycle if ℓ splits in $\mathcal{O}$ and $n \geq 3$, where n is the order of $[\mathfrak{a}]$



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The inverse volcano problem

Question

Fix $k \geq 1$. Given an abstract ℓ -volcano V , can we find primes $p \neq \ell$ such that V is a connected component of $\mathcal{G}_\ell(\mathbb{F}_{p^k})$?

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Figure: The volcano $(S_2, 2, 1)$.

The abstract 2-volcano $V = (S_2, 2, 1)$ is **impossible** to find in any $\mathcal{G}_2(\mathbb{F}_{p^2})$!

Our results thus far

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Prime fields

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Depth zero, non-prime fields

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$$d = 0, k > 1$$

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Ramified principal, depth non-zero,
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Ramified non-principal, depth non-zero, non-prime fields

$$V_0 = R_2, d > 0, k > 1, \ell \geq 5$$

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Existence of a volcano

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Note: When $d = 0$, things are easier as the volcanoes are just craters which don't depend on ℓ . So we are free to choose ℓ .

The strategy for depth

Lemma

Let $\mathcal{O}$ be an order and L its ring class field.

Unramified primes q above p in L = ordinary elliptic curves with CM by $\mathcal{O}$ existing over characteristic p .

Residue degree of q = degree of the field extension = order of σ_q in $\text{Gal}(L/K)$.

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If we want a volcano of depth d we need...

- ▶ an element in $\text{Gal}(L_d/K)$ with order dividing k ,
- ▶ such that its pre-image in $\text{Gal}(L_{d+1}/K)$ has order not dividing k .

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Call the set of such elements X . Chebotarev's density theorem then tells us that the primes of K unramified in L_{d+1} whose Frobenius element is in X has density $|X|/h(\mathcal{O}_{d+1})$.

The strategy for depth

Theorem

Let $\mathcal{O}_0$ be an order of discriminant $D < -4$ with conductor c in an imaginary quadratic field $K = \mathbb{Q}(\sqrt{D})$. Fix a prime ℓ coprime to c . Let $\mathcal{L}$ be a prime ideal in $\mathcal{O}_0$ lying above ℓ and let V_0 be the abstract crater associated with $\mathcal{L}$. Fix $k \geq 1$ and $d \geq 0$.

There exists a prime $p = p(k, d) \in \mathbb{Z}^+$ such that $\mathcal{G}_\ell(\mathbb{F}_{p^k})$ contains the volcano (V_0, ℓ, d) as a connected component if and only if the set

$$X = \{\sigma \in \text{Gal}(L_{d+1}/K) \mid \text{ord}(\sigma) \nmid k \text{ and } \text{ord}(\pi_d(\sigma)) \mid k\},$$

is non-empty.

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If such a prime p exists, then there are in fact infinitely many of them.

- ▶ Only need one order for existence.
- ▶ But need to rule out *every* order for non-existence.

The strategy in action: prime fields

Lemma

When $k = 1$, $|X| = 0$ if and only if $\ell = 2$ splits in $\mathcal{O}_K$ and $d = 0$.

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Proof. Using the class number formula:

$$|X| = |\ker \pi_d| - 1 = |\mathrm{Gal}(L_{d+1}/L_d)| - 1 = \begin{cases} \ell - (D/\ell) - 1 & \text{if } d = 0, \\ \ell - 1 & \text{otherwise.} \end{cases} \quad \blacksquare$$

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What about when $\ell = 2$ splits in $\mathcal{O}_K$ and $d = 0$? Remember we are free to choose ℓ when $d = 0$, so just don't pick 2!

What about the crater orders?

Bambury, Campagna, and Pazuki [5] construct orders for every prime ℓ and crater V_0 such that a prime ideal lying above ℓ realises V_0 .

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Example

Let $K = \mathbb{Q}(\sqrt{1 - 2^{n+2}})$ and $n \neq 4$ a positive integer. Then in $\mathcal{O}_K$ the prime 2 splits into two prime ideals whose corresponding classes in $\text{Cl}(\mathcal{O}_K)$ have order n .

This is the crater S_n for $n \neq 4$ when $\ell = 2$.

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Bambury, Campagna, and Pazuki [5] construct orders for every prime ℓ and crater V_0 such that a prime ideal lying above ℓ realises V_0 .

Example

Let $K = \mathbb{Q}(\sqrt{1 - 2^{n+2}})$ and $n \neq 4$ a positive integer. Then in $\mathcal{O}_K$ the prime 2 splits into two prime ideals whose corresponding classes in $\text{Cl}(\mathcal{O}_K)$ have order n .

This is the crater S_n for $n \neq 4$ when $\ell = 2$.

Combining this with our previous lemma and theorem, we have solved the inverse volcano problem over $\mathbb{F}_p$: for any volcano V we can always find a prime p (in fact, infinitely many of them!) such that V is in $\mathcal{G}_\ell(\mathbb{F}_p)$.



Thank you for listening!

References

- [1] E. Z. Goren and J. R. Love, “Supersingular elliptic curves, quaternion algebras and applications to cryptography,” Oct. 2024, doi: [10.48550/arXiv.2410.06123](https://doi.org/10.48550/arXiv.2410.06123).
- [2] A. V. Sutherland, “Isogeny volcanoes,” *The Open Book Series*, vol. 1, no. 1, pp. 507–530, Nov. 2013, doi: [10.2140/obs.2013.1.507](https://doi.org/10.2140/obs.2013.1.507).
- [3] D. R. Kohel, “Endomorphism rings of elliptic curves over finite fields,” 1996.
- [4] M. Fouquet and F. Morain, “Isogeny volcanoes and the SEA algorithm,” in *Algorithmic Number Theory (Sydney, 2002)*, C. Fieker and D. R. Kohel, Eds., Lecture Notes in Computer Science: Springer, Berlin, 2002, pp. 276–291. doi: [10.1007/3-540-45455-1_23](https://doi.org/10.1007/3-540-45455-1_23).
- [5] H. Bambury, F. Campagna, and F. Pazuki, “Ordinary isogeny graphs over $\mathbb{F}_p$: the inverse volcano problem”, *Annali della Scuola Normale Superiore di Pisa. Classe di Scienze. Serie V*, p. 33, July 2024, doi: [10.2422/2036-2145.202310_022](https://doi.org/10.2422/2036-2145.202310_022).